

Module 4: Bayesian Games, Signaling, and Verifiable Disclosure

Exam-Facing Lecture Notes for the ASU Microeconomics PhD Qual

PhD Qual Lecture Notes Builder

Built from ASU Bayesian-games notes, homeworks, Maschler–Solan–Zamir, and past qualifying exams

Contents

How to use this module	3
1 Source inventory and exam map	3
1.1 Primary sources used	3
1.2 Past-qual inventory: what has actually been asked	4
2 Detailed blueprint	5
2.1 Learning objectives	5
2.2 Exam sequence	5
2.3 Canonical model coverage	6
3 The ASU four-layer answer protocol	6
3.1 The exact answer frame	6
3.2 Symbol discipline	7
4 Bayesian games and Bayesian Nash equilibrium	7
4.1 Canonical Bayesian game	7
4.2 Pure-BNE algorithm for finite type/action problems	8
4.3 Canonical computation: two-type R&D race	9
4.4 Bayesian Cournot with private costs	9
5 Assessments, WPBE, and sequential equilibrium	10
5.1 Assessments and beliefs	10
5.2 WPBE verification in signaling games	11
5.3 WPBE versus sequential equilibrium	12
6 Spence job-market signaling	12
6.1 Two-type Spence model	12
6.2 Pooling equilibria	13
6.3 Finite-type separating recursion	13
6.4 Continuum-type differential equation	14

7	Screening, adverse selection, and single crossing	15
7.1	Direct screening menu	15
7.2	Rothschild–Stiglitz insurance screening	16
7.3	Credit-market screening	17
8	Verifiable disclosure and hard information	17
8.1	Canonical verifiable-disclosure game	18
8.2	Salesperson quality disclosure	18
8.3	Disclosure in Bayesian Cournot	20
9	Graphical tools	20
9.1	Wage-education diagrams	20
9.2	Disclosure lattice	21
9.3	Insurance state-space diagrams	21
9.4	Bayesian best-reply diagrams	21
10	Proof-writing templates	22
10.1	Template: prove a candidate is BNE	22
10.2	Template: prove a candidate is WPBE	22
10.3	Template: prove monotonicity from single crossing	22
10.4	Template: prove no pooling in disclosure	23
11	Qual-style drills with solution skeletons	23
11.1	Drill 1: enumerate pure BNE in a two-type game	23
11.2	Drill 2: find a separating Spence equilibrium	23
11.3	Drill 3: determine whether pooling on an ambiguous disclosure is wPBE	23
11.4	Drill 4: write a screening menu problem	23
12	Common pitfalls and fixes	24
12.1	Pitfall: forgetting receiver strategies off path	24
12.2	Pitfall: treating BNE as state-by-state Nash equilibrium	24
12.3	Pitfall: using FOCs before proving concavity or monotonicity	24
12.4	Pitfall: saying “pooling” when the receiver still learns	24
13	One-page formula sheet	24
13.1	Bayesian games	25
13.2	WPBE	25
13.3	Spence two-type separation	25
13.4	Screening monotonicity	25
13.5	Disclosure unraveling	25
14	References for further study	25

How to use this module

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw one page with Nature at the top, information sets in the middle, and payoff/equilibrium checks at the bottom.* This module is written for exam production, not for passive reading. Every canonical model is organized in the required four-layer answer frame

Object $\rightarrow$ Feasibility $\rightarrow$ Optimality/Equilibrium $\rightarrow$ Verification

Use the same four headings in your answer booklet whenever the question asks you to “set up,” “define equilibrium,” “identify all equilibria,” or “prove that the candidate is an equilibrium.” The format forces you to specify who knows what, what strategies are functions of, which constraints must hold, and how beliefs or deviations are checked.

Exam tip. For ASU-style micro questions, a correct answer often gets most of its value before the algebra starts: name the primitive objects, specify the strategy spaces, write the belief system, and state which inequalities verify equilibrium. Algebra without the object and feasibility layer is fragile.

1 Source inventory and exam map

Visual intuition hook. *Pair with Visual Intuition Curriculum: make a two-column map: left side lists lecture sources, right side lists past qual problem types and the skill each tests.*

1.1 Primary sources used

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a source-flow diagram from notes to canonical models to past-qual drills.* The source folder for Bayesian Games contains the core notes used here: `dgt015-bgames.tex`, `dgt018-sp.tex`, `dgt019-th.tex`, `dgt020-wpbe.tex`, `dgt021-se.tex`, `dgt022-fwdi.tex`, `dgt024-sgex.tex`, `die026-advsel.tex`, `die027-spence.tex`, `die028-rs.tex`, and `die030-pagent.tex`. The repository README classifies the Bayesian-games block as equilibrium refinements, auctions, signaling, screening, and principal-agent material, and indicates that homeworks and old exams are practice sources. These notes use that taxonomy but focus Module 4 on Bayesian-game formulation, signaling/WPBE/SE, Spence signaling, screening as the dual problem, and verifiable disclosure.

Source	How it is used in this module
<code>dgt015-bgames.tex</code>	Harsanyi transformation, Bayesian game primitives, common priors, pure/behavior strategies, BNE definitions, finite-game existence, IPV notation.
<code>dgt020-wpbe.tex</code>	Assessments, beliefs at information sets, WPBE, Bayes-rule restrictions, sequential rationality checks, examples in which NE/SPE are too weak.
<code>dgt021-se.tex</code>	Consistency of beliefs, sequential equilibrium, fully mixed perturbations, SE existence in finite perfect-recall games.
<code>dgt022-fwdi.tex</code>	Forward-induction intuition and equilibrium dominance; used here as refinement intuition for costly signaling and disclosure.

dgt024-sgex.tex	Signaling-game definition, separating and pooling terminology, infinite-signal caveats.
die026-advsel.tex	Adverse selection, lemons logic, market breakdown, and why signaling/screening arise.
die027-spence.tex	Two-type Spence job-market signaling, zero-profit wage schedules, separating/pooling IC conditions.
die028-rs.tex	Rothschild–Stiglitz insurance screening diagrams, zero-profit lines, full insurance for high risk and distortion for low risk.
Maschler–Solán–Zamir, <i>Game Theory</i>	Equilibrium refinements, games with incomplete information and common priors, universal belief spaces, and auctions; especially useful for proving WPBE/SE claims cleanly.
Homeworks	Signaling and principal-agent problem sets provide additional practice in writing IC/IR systems and interpreting off-path beliefs.
Past ASU quals	Used to select canonical computations and proof templates.

1.2 Past-qual inventory: what has actually been asked

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a calendar heat map with BNE, signaling, screening, and disclosure as rows.* The table below is an exam-facing inventory. “Setup” means the exam has repeatedly rewarded exact strategy spaces and belief systems. “Computation” means the problem requires solving inequalities or best replies. “Proof” means the problem asks for theorem-style verification, often with monotonicity or equilibrium-refinement logic.

Exam	Topic	Skill tested	Tag
Aug. 2018 Q1	Two-type R&D race	Pure strategies as type-contingent maps; pure BNE enumeration; robustness of weak equilibria.	BNE
June 2018 Q3	Three-type signaling game	Pure strategies for sender and receiver; WPBE definition; separating vs pooling WPBE.	WPBE
June 2021 Q4	Cournot with private costs and disclosure message	Bayesian Cournot best replies; assessments after messages; whether pooling on ambiguous message can be wPBE.	disclosure
June 2022 Q4	Bayesian Cournot disclosure game	BNE, complete-information benchmark, profitable deviation by low-cost type from silence; no nonlearning wPBE.	disclosure
June 2024 Q3	Verifiable salesperson messages	Buyer demand from posterior mean quality; BNE vs wPBE; all wPBE; welfare comparison with no communication.	disclosure
Aug. 2022 Q3	Two-type Spence training	Define WPBE; prove wages equal type value in separation; derive interval of separating training levels and least-cost separation.	signaling
Aug. 2023 Q3	Finite-type Spence recursion	Single crossing; construct separating education levels recursively.	signaling

Aug. 2024 Q3	Numerical education-threshold signaling	Indifference curves, complete-information signaling benchmark, separating thresholds, pooling under average wage.
June 2023 Q4	Monopoly menu with finite types	SSCP/increasing differences equivalence, screening monotone choice theorem, adjacent IC/IR logic.
June 2019 Q4	Competitive credit screening	Observable-type zero collateral benchmark; unique separating candidate; existence/nonexistence logic.
June 2020 Q2	Competitive insurance with adverse selection	Equilibrium definition, existence, market breakdown/trade, two-type screening equations and diagram.

2 Detailed blueprint

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a subway map whose stops are: Harsanyi model, BNE, assessment, WPBE, SE, Spence, screening, disclosure.*

2.1 Learning objectives

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw the same signaling tree three times: with only strategies, then with beliefs, then with deviation arrows.* By the end of this module you should be able to:

1. formulate a Bayesian game from words: players, types, actions, priors, payoff functions, strategies, and beliefs;
2. write pure and behavioral BNE conditions both type-by-type and ex ante;
3. compute pure-strategy BNE in finite type/action games by constructing type-by-type best replies;
4. define assessment, WPBE, consistency, and sequential equilibrium in finite signaling games;
5. verify separating and pooling equilibria by checking receiver optimality, Bayes' rule, and sender deviations;
6. derive Spence separating intervals, least-cost separating equilibria, and pooling conditions;
7. draw and use graphical tools: wage-education diagrams, belief trees, insurance state-space diagrams, and disclosure/unraveling lattices;
8. prove monotonicity from single crossing and use it to simplify screening and signaling problems;
9. solve verifiable-disclosure problems where a type can tell the truth or stay ambiguous but cannot lie;
10. write short, rigorous proof paragraphs that satisfy the Object–Feasibility–Optimality/Equilibrium–Verification frame.

2.2 Exam sequence

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw four stacked boxes labeled primitives, constraints, equilibrium inequalities, and final verification.* A reliable exam sequence is:

1. **Translate words into primitives.** Write the type space before the action space. Specify who observes the type.

2. **Write strategy sets as functions.** If a player has private information, her pure strategy maps own type to action. If a receiver acts after a message, her strategy maps each message to an action.
3. **Write beliefs at every information set.** On path, Bayes' rule pins them down. Off path, WPBE permits free beliefs; SE restricts them by perturbations.
4. **Compute continuation optimality.** Receiver first, then sender type-by-type. In simultaneous Bayesian games, compute all type-contingent best replies.
5. **Do not stop at a candidate.** Say why no other equilibrium exists: exhaust cases, use monotonicity, or identify a profitable deviation.

2.3 Canonical model coverage

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a matrix with rows as models and columns as Object, Feasibility, Equilibrium, Verification.* The module has six canonical model blocks:

1. Bayesian normal-form games and BNE.
2. Assessments, WPBE, and sequential equilibrium.
3. Spence job-market signaling.
4. Screening and single-crossing menus.
5. Rothschild–Stiglitz insurance/credit screening.
6. Verifiable disclosure and unraveling.

Each block begins with the four-layer answer frame and ends with a verification checklist.

3 The ASU four-layer answer protocol

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a proof as a flowchart: object definitions feed constraints, constraints feed inequalities, inequalities feed verification.*

3.1 The exact answer frame

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw four horizontal lanes and place each symbol in exactly one lane.* Use this form whenever possible:

Object. State the primitive mathematical object: players, type spaces, action/message spaces, timing, information, payoff functions, priors, and the equilibrium concept.

Feasibility. State all constraints: strategy functions, belief restrictions, Bayes' rule where applicable, IC/IR, zero-profit, resource, budget, or message-feasibility constraints.

Optimality/Equilibrium. Write the optimality or equilibrium inequalities: best replies, sequential rationality, sender IC, receiver posterior-optimal action, firm zero profit, or principal's objective.

Verification. Close the proof: check all deviations, all types, all messages/information sets, all constraints, and then show uniqueness or characterize multiplicity.

Proof-writing template. A compact exam paragraph can be as simple as: “Object: A pure strategy for type-informed player 1 is $s_1 : T_1 \rightarrow A_1$ and for receiver 2 is $s_2 : M \rightarrow A_2$. Feasibility: beliefs $\mu(\cdot | m)$ must satisfy Bayes’ rule whenever m is reached. Equilibrium: for each type t , $s_1(t)$ maximizes $u_1(t, m, s_2(m))$, and for each message m , $s_2(m)$ maximizes posterior expected u_2 . Verification: compute posteriors, compare every sender deviation, compare every receiver action, and state whether off-path beliefs can be chosen to support the candidate.”

3.2 Symbol discipline

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a notation ledger with separate columns for type, message, action, belief, payoff. Keep notation separate:*

- t_i or θ_i : type/private information.
- a_i or q_i : simultaneous action/output.
- m or e : message/signal/training/education observed by a receiver.
- s_i : pure strategy, usually a function.
- σ_i : mixed or behavioral strategy.
- $\mu(\cdot | h)$ or $\mu(\cdot | m)$: belief at information set or after message.
- u_i or U_i : payoff/expected utility.

Common mistake. Never write “player 1 chooses type” unless the model says she does. Nature chooses types; players choose actions conditional on their information. In Bayesian games, strategies are functions from types to actions, not just actions.

4 Bayesian games and Bayesian Nash equilibrium

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw Nature choosing a type vector, then each player observing only her component before simultaneous actions.*

4.1 Canonical Bayesian game

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a Harsanyi transformation: an incomplete-information normal-form problem becomes Nature plus type-contingent strategies.*

Object. A finite Bayesian game is

$$\Gamma = (N, \{T_i\}_{i \in N}, \{A_i\}_{i \in N}, P, \{u_i\}_{i \in N}),$$

where N is the finite player set, T_i is player i ’s type set, A_i is player i ’s action set, $P \in \Delta(\prod_i T_i)$ is the common prior, and

$$u_i : T \times A \rightarrow \mathbb{R}, \quad T = \prod_i T_i, \quad A = \prod_i A_i.$$

Player i observes t_i and not necessarily t_{-i} .

Feasibility. A pure strategy is a function

$$s_i : T_i \rightarrow A_i.$$

A behavioral strategy is a function

$$\sigma_i : T_i \rightarrow \Delta(A_i).$$

If $P(t_i) > 0$, player i 's conditional belief is

$$\beta_i(t_{-i} | t_i) = \frac{P(t_i, t_{-i})}{\sum_{\hat{t}_{-i}} P(t_i, \hat{t}_{-i})}.$$

If $P(t_i) = 0$, Bayes' rule does not pin down $\beta_i(\cdot | t_i)$; most exam problems avoid zero-probability types or explicitly tell you how to treat them.

Optimality/Equilibrium. A pure strategy profile s^* is a BNE if for every player i , every type t_i with positive probability, and every action $a_i \in A_i$,

$$\sum_{t_{-i}} u_i(t_i, t_{-i}, s_i^*(t_i), s_{-i}^*(t_{-i})) \beta_i(t_{-i} | t_i) \geq \sum_{t_{-i}} u_i(t_i, t_{-i}, a_i, s_{-i}^*(t_{-i})) \beta_i(t_{-i} | t_i).$$

Equivalently, each type-agent (i, t_i) is best responding in the agent-normal form.

Verification. To verify a BNE, fix one player and one type, hold all other players' strategies fixed, compute expected payoff from each feasible action, and check that the prescribed action is maximal. Repeat for every type of every player. To show all pure BNE, list all type-contingent strategies and build best-response correspondences.

Definition 4.1 (Agent-normal form). For each pair (i, t_i) create an agent who chooses an action in A_i . Expected payoff to agent (i, t_i) is player i 's conditional expected payoff given t_i . A pure BNE of the Bayesian game is a pure NE of this agent-normal-form game.

Theorem 4.2 (Finite BNE existence). *Every finite Bayesian game has a mixed or behavioral BNE.*

Proof. The agent-normal form is a finite normal-form game. Nash's existence theorem gives a mixed-strategy NE. In a finite Bayesian game, a mixed strategy over type-contingent pure strategies induces a behavioral strategy at each type; with perfect recall, the induced strategic behavior is payoff equivalent for the Bayesian normal-form problem. Hence a BNE exists. $\square$

4.2 Pure-BNE algorithm for finite type/action problems

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a table with opponent type-contingent strategies as columns and own-type best replies as rows.*

1. List pure strategies as complete contingent plans. With two types and two actions, there are four pure strategies.
2. For each opponent strategy profile, compute each type's expected payoff from each action.
3. Convert type-by-type best actions into a set of best pure strategies.
4. Intersect best-response correspondences across players.
5. Flag weak equilibria that rely on indifference or weakly dominated type actions.

Common mistake. Do not average over your own type when checking a type's incentive constraint. A type knows who she is. Ex ante payoff inequalities can be equivalent only when all own types have positive probability and the deviation can be chosen type by type.

4.3 Canonical computation: two-type R&D race

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a 2×2 type grid and shade the cells in which both firms enter and the race occurs.*

Object. Two firms $i = 1, 2$ have independent types $t_i \in \{s, w\}$, each with probability $1/2$. Each firm chooses Y or N . A race occurs only if both choose Y . If both enter, each pays cost 2; a winner gets 10. Strong beats weak; equal types split the winning probability. A pure strategy is a map $\{s, w\} \rightarrow \{Y, N\}$, written (a_s, a_w) .

Feasibility. The four pure strategies are

$$YY = (Y, Y), \quad YN = (Y, N), \quad NY = (N, Y), \quad NN = (N, N).$$

If a firm does not enter, or if the other firm does not enter, its payoff is 0. If both enter, payoffs from entering are:

$$\pi_s(Y | s) = 3, \quad \pi_s(Y | w) = 8, \quad \pi_w(Y | s) = -2, \quad \pi_w(Y | w) = 3.$$

Optimality/Equilibrium. Against an opponent strategy $(y_s, y_w) \in \{0, 1\}^2$, where 1 means enter,

$$U_s(Y) = \frac{1}{2}(3y_s + 8y_w), \quad U_w(Y) = \frac{1}{2}(-2y_s + 3y_w),$$

and $U_s(N) = U_w(N) = 0$. Thus

$$\text{BR}(YY) = \{YY\}, \quad \text{BR}(YN) = \{YN\}, \quad \text{BR}(NY) = \{YY\}, \quad \text{BR}(NN) = \{YY, YN, NY, NN\}.$$

Verification. A pure-strategy BNE is a pair (s_1, s_2) with $s_i \in \text{BR}(s_j)$ for both players. Therefore the pure BNE are

$$(YY, YY), \quad (YN, YN), \quad (NN, NN).$$

The profile (NY, NY) is not a BNE: if the opponent's weak type enters, a strong type strictly prefers Y to N . The equilibrium (NN, NN) is weak and nonrobust because when the opponent never enters both types are indifferent, and the strong type's action Y weakly dominates N .

4.4 Bayesian Cournot with private costs

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw best replies in output space, then add a second line for each cost type.*

Object. There are two firms with inverse demand $P(Q) = A - Q$. Firm 1 has privately known unit cost $k + \theta$, where $\theta \in \{\theta_L, \theta_H\}$, $\theta_L < 0 < \theta_H$, each with probability $1/2$ and $\mathbb{E}\theta = 0$. Firm 2 has known cost k . Firm 1 observes θ ; firm 2 observes only the prior.

Feasibility. A pure strategy for firm 1 is $q_1 : \{\theta_L, \theta_H\} \rightarrow \mathbb{R}_+$; write $q_L = q_1(\theta_L)$ and $q_H = q_1(\theta_H)$. A pure strategy for firm 2 is a number $q_2 \in \mathbb{R}_+$.

Optimality/Equilibrium. Assuming interior outputs, firm 1 type θ solves

$$\max_{q \geq 0} (A - q - q_2 - k - \theta)q, \quad \text{so} \quad q_\theta = \frac{A - k - \theta - q_2}{2}.$$

Firm 2 solves

$$\max_{q \geq 0} \mathbb{E}[(A - q_1(\theta) - q - k)q], \quad \text{so} \quad q_2 = \frac{A - k - \mathbb{E}q_1(\theta)}{2}.$$

Using $\mathbb{E}\theta = 0$ gives

$$\mathbb{E}q_1(\theta) = \frac{A - k - q_2}{2}, \quad q_2 = \frac{A - k}{3}, \quad q_\theta = \frac{A - k}{3} - \frac{\theta}{2}.$$

Verification. Check nonnegativity under the parameter restrictions. Then verify each FOC is sufficient because each objective is strictly concave in own output over the relevant region. If boundary cases are possible, replace each formula by the positive-part best reply

$$q_\theta \in \arg \max_{q \geq 0} (A - q - q_2 - k - \theta)q, \quad q_2 \in \arg \max_{q \geq 0} \mathbb{E}[(A - q_1(\theta) - q - k)q].$$

5 Assessments, WPBE, and sequential equilibrium

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a signaling tree and write one posterior next to every receiver information set.*

5.1 Assessments and beliefs

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw an information set as a dashed oval containing two nodes with probabilities summing to one.*

Object. A finite extensive-form game with imperfect information has histories/nodes, information sets, behavioral strategies, and payoff functions. An assessment is a pair (σ, μ) consisting of a behavioral-strategy profile σ and a system of beliefs μ .

Feasibility. For every player i and every information set $h \in \mathcal{H}_i$, $\mu_i(\cdot \mid h)$ is a probability distribution over nodes in h . If h is reached with positive probability under σ , then Bayes' rule requires

$$\mu_i(x \mid h) = \frac{P_\sigma(x)}{P_\sigma(h)} \quad (x \in h).$$

If $P_\sigma(h) = 0$, WPBE leaves beliefs unrestricted; sequential equilibrium does not.

Optimality/Equilibrium. Sequential rationality requires that at each information set $h \in \mathcal{H}_i$, player i 's continuation strategy maximizes expected continuation payoff given $\mu_i(\cdot \mid h)$ and the continuation strategies of other players:

$$E[u_i \mid h, \mu_i, \sigma_i, \sigma_{-i}] \geq E[u_i \mid h, \mu_i, \sigma'_i, \sigma_{-i}] \quad \text{for all } \sigma'_i.$$

Verification. To verify a WPBE, check: (i) beliefs are probability distributions; (ii) Bayes' rule holds at every reached information set; (iii) each player is sequentially rational at every information set. To verify a sequential equilibrium, also construct or argue existence of completely mixed $\sigma^n \rightarrow \sigma$ whose Bayes-rule beliefs μ^n converge to μ .

Definition 5.1 (WPBE). An assessment (σ, μ) is a weak perfect Bayesian equilibrium if it satisfies sequential rationality and Bayes' rule wherever possible.

Definition 5.2 (Sequential equilibrium). An assessment (σ, μ) is a sequential equilibrium if it is sequentially rational and consistent: there exists a sequence of completely mixed behavior strategies $\sigma^n \rightarrow \sigma$ such that the Bayes-rule beliefs μ^n induced by σ^n satisfy $\mu^n \rightarrow \mu$.

Theorem 5.3 (Existence of sequential equilibrium). *Every finite extensive-form game with perfect recall has a sequential equilibrium.*

5.2 WPBE verification in signaling games

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw two arrows from a sender type to messages and a receiver best-response table below each message.*

Object. In a signaling game, Nature draws sender type $t \in T$ from prior P . Sender observes t and sends message/signal $m \in M(t)$, where $M(t)$ may depend on type. Receiver observes m , forms belief $\mu(\cdot | m)$, and chooses action $a \in A$.

Feasibility. Sender strategy $s_1 : T \rightarrow M$ must satisfy $s_1(t) \in M(t)$. Receiver strategy $s_2 : M \rightarrow A$ assigns an action after every message. Beliefs satisfy

$$\mu(t | m) = \frac{P(t)\mathbf{1}\{s_1(t) = m\}}{\sum_{\hat{t}} P(\hat{t})\mathbf{1}\{s_1(\hat{t}) = m\}}$$

when the denominator is positive.

Optimality/Equilibrium. Receiver optimality requires

$$s_2(m) \in \arg \max_{a \in A} \sum_t \mu(t | m) u_2(t, m, a).$$

Sender optimality requires, for every type t and every feasible deviation $m' \in M(t)$,

$$u_1(t, s_1(t), s_2(s_1(t))) \geq u_1(t, m', s_2(m')).$$

Verification. The fastest verification order is receiver first, sender second. First compute posteriors and receiver best replies after every message. Then check sender deviations type by type. Finally, explain whether off-path beliefs can support any claimed off-path receiver action. For SE, add the perturbation or explain why Bayes' rule along every approximating sequence forces a different posterior.

Proof-writing template. When asked to “find all WPBE,” classify sender strategies by the partition they induce on types: separating, pooling, and semi-pooling. For each partition, Bayes' rule pins down on-path beliefs. Compute receiver actions. Then use sender IC inequalities to keep or discard the partition.

5.3 WPBE versus sequential equilibrium

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw an off-path information set with a free WPBE belief and then a limiting belief from trembles.* WPBE can allow implausible off-path beliefs. Sequential equilibrium restricts beliefs to those obtainable as limits from completely mixed strategies. In exams, the typical task is not to master all refinements but to state exactly why a belief is unrestricted under WPBE and why it may be restricted under SE.

Example 5.4 (A consistency check). Suppose an information set h is reached only if sender chooses R , and both types have the same small tremble probability x_n of choosing R under any completely mixed sequence converging to a pooling-on- L strategy. If the prior is $(1/2, 1/2)$, Bayes' rule gives

$$\mu^n(H | h) = \frac{(1/2)x_n}{(1/2)x_n + (1/2)x_n} = \frac{1}{2}.$$

Thus any claimed sequential equilibrium with $\mu(H | h) \neq 1/2$ fails consistency, even if it is a WPBE.

6 Spence job-market signaling

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw education on the horizontal axis and wage on the vertical axis; high types have flatter indifference curves.*

6.1 Two-type Spence model

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw two indifference curves crossing once, with the high type's curve flatter in (e, w) space.*

Object. Nature chooses worker productivity $\eta \in \{\eta_L, \eta_H\}$, with $0 < \eta_L < \eta_H$ and $\mathbb{P}(\eta_H) = \lambda$. The worker observes η and chooses education $e \geq 0$. Education has no direct productivity value but costs $c(e, \eta)$ to the worker. Firms observe e and offer a wage $w(e)$. Worker payoff is

$$U_W(\eta, e, w) = w - c(e, \eta),$$

and a firm hiring type η at wage w receives $\eta - w$.

Feasibility. A worker strategy is $e : \{\eta_L, \eta_H\} \rightarrow \mathbb{R}_+$. A market wage schedule is $w : \mathbb{R}_+ \rightarrow \mathbb{R}$. A belief system assigns $\mu(\eta_H | e)$ after every education level. In a competitive market, zero expected profit implies

$$w(e) = \mathbb{E}[\eta | e] = \mu(\eta_H | e)\eta_H + (1 - \mu(\eta_H | e))\eta_L.$$

Optimality/Equilibrium. In a separating equilibrium with $e_L = 0$ and $e_H = e^* > 0$, Bayes' rule gives $w(0) = \eta_L$ and $w(e^*) = \eta_H$. Sender IC requires

$$\eta_L - c(0, \eta_L) \geq \eta_H - c(e^*, \eta_L),$$

$$\eta_H - c(e^*, \eta_H) \geq \eta_L - c(0, \eta_H).$$

If $c(0, \eta) = 0$, these become

$$c(e^*, \eta_L) \geq \Delta\eta, \quad c(e^*, \eta_H) \leq \Delta\eta, \quad \Delta\eta = \eta_H - \eta_L.$$

Verification. A separating education level is feasible if it satisfies both IC inequalities and if off-path beliefs make no type want an unassigned education. The least-cost separating equilibrium sets the low type just indifferent:

$$e^* = c_L^{-1}(\Delta\eta) \quad \text{when } c_L(e) = c(e, \eta_L),$$

provided $c_H(e^*) \leq \Delta\eta$. Refinements such as the intuitive criterion usually select this least-cost separating signal and eliminate many pooling equilibria.

6.2 Pooling equilibria

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw both types choosing the same education point and one average wage above it.*

Object. A pooling equilibrium has $e_L = e_H = \bar{e}$ and on-path wage

$$\bar{w} = (1 - \lambda)\eta_L + \lambda\eta_H.$$

Feasibility. Bayes' rule pins down the on-path belief at $\bar{e}$ to the prior. Off-path beliefs are free under WPBE. The receiver wage after every off-path education must equal expected productivity under the chosen off-path belief if the market is competitive.

Optimality/Equilibrium. Type η must prefer pooling to every off-path deviation e' :

$$\bar{w} - c(\bar{e}, \eta) \geq w(e') - c(e', \eta) \quad \forall e' \geq 0.$$

A common construction sets off-path beliefs to “low type” after deviations, giving $w(e') = \eta_L$ for $e' \neq \bar{e}$. Then check both types against the best deviation to $e' = 0$ or to a nearby signal.

Verification. Pooling equilibria are belief-sensitive. A WPBE may exist under pessimistic off-path beliefs, but a refinement can remove it if the high type would gain from a deviation that the low type would not rationally take. Always state the off-path belief and the exact IC inequality it supports.

6.3 Finite-type separating recursion

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a staircase of education levels, one step for each higher type.*

Object. Types are $x_1 < x_2 < \dots < x_N$, with $x_n - x_{n-1} = 2$. Type x_n has productivity x_n and payoff

$$u(x_n, e, w) = w - k(x_n)e^2,$$

where $k(x) > 0$ is decreasing in x . Firms pay expected productivity conditional on education.

Feasibility. A separating strategy assigns distinct education levels $e_n = e(x_n)$. Competition gives wage $w(e_n) = x_n$. Set $e_1 = 0$ for least-cost separation.

Optimality/Equilibrium. The adjacent low type's no-mimic constraint is

$$x_{n-1} - k(x_{n-1})e_{n-1}^2 \geq x_n - k(x_{n-1})e_n^2.$$

For least-cost separation bind this constraint:

$$k(x_{n-1})(e_n^2 - e_{n-1}^2) = x_n - x_{n-1} = 2,$$

so

$$\boxed{e_n^2 = e_{n-1}^2 + \frac{2}{k(x_{n-1})}}, \quad e_1 = 0.$$

Verification. The higher type does not want to move down because

$$x_n - k(x_n)e_n^2 \geq x_{n-1} - k(x_n)e_{n-1}^2 \iff k(x_n)(e_n^2 - e_{n-1}^2) \leq 2,$$

which follows from $k(x_n) < k(x_{n-1})$ and the recursion. Non-adjacent deviations are ruled out by summing adjacent IC inequalities and single crossing.

6.4 Continuum-type differential equation

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a smooth wage curve tangent to each type's indifference curve.*

Object. Type $t \in [0, 1]$ has firm value $v(t) = \sqrt{t}$ and education cost e/t for $t > 0$. Worker payoff is $w(e) - e/t$. In a differentiable separating equilibrium, education $e(t)$ is strictly increasing and the market wage satisfies

$$w(e(t)) = \sqrt{t}.$$

Feasibility. Differentiability and separation let us invert $e(t)$ and write $t = t(e)$. Zero profit gives $w(e) = \sqrt{t(e)}$.

Optimality/Equilibrium. The worker's FOC at $e(t)$ is

$$w'(e(t)) = \frac{1}{t}.$$

Since $w(e(t)) = \sqrt{t}$, differentiating gives

$$w'(e(t))e'(t) = \frac{1}{2\sqrt{t}}.$$

Thus

$$e'(t) = \frac{t}{2\sqrt{t}} = \frac{\sqrt{t}}{2}, \quad \text{or equivalently} \quad \frac{dw}{de} = \frac{1}{w^2}.$$

Solving the wage ODE gives $w(e)^3 = 3e + C$. With $w(0) = 0$, $w(e) = (3e)^{1/3}$.

Verification. If $w(e) = (3e)^{1/3}$, type t in separation receives $w = \sqrt{t}$, so

$$e(t) = \frac{t^{3/2}}{3}, \quad \text{education cost} \quad \frac{e(t)}{t} = \frac{\sqrt{t}}{3}.$$

Thus one-third of firm value $\sqrt{t}$ is spent on education costs. Check the second-order/single-crossing condition to ensure the FOC characterizes the global choice.

7 Screening, adverse selection, and single crossing

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a menu of contracts as dots, with each type choosing a different dot.*

7.1 Direct screening menu

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw type on the vertical axis and contracts ordered northeast in (q, T) space.*

Object. A monopolist offers contracts $(q, T) \in \mathbb{R}_+^2$ to a buyer with type $\theta \in \{\theta_1 < \dots < \theta_n\}$. Type θ has quasilinear utility

$$u(q, T, \theta) = f(q, \theta) - T,$$

where f is increasing in q . The outside option is $(0, 0)$.

Feasibility. A menu $\{(q_i, T_i)\}_{i=1}^n$ is incentive compatible and individually rational if, for every i, j ,

$$f(q_i, \theta_i) - T_i \geq f(q_j, \theta_i) - T_j \quad (\text{IC})$$

$$f(q_i, \theta_i) - T_i \geq 0 \quad (\text{IR}).$$

The revelation principle permits restricting attention to direct menus indexed by reported type.

Optimality/Equilibrium. If f has strictly increasing differences in (q, θ) , then higher types have a stronger marginal willingness to pay for q . Profit maximization chooses feasible (q_i, T_i) to maximize

$$\sum_i p_i T_i$$

subject to IC and IR, when production cost is zero. With two types, the usual binding constraints are low-type IR and high-type downward IC:

$$f(q_1, \theta_1) - T_1 = 0,$$

$$f(q_2, \theta_2) - T_2 = f(q_1, \theta_2) - T_1.$$

Verification. To verify monotonicity, combine IC for $i < j$:

$$f(q_i, \theta_i) - T_i \geq f(q_j, \theta_i) - T_j,$$

$$f(q_j, \theta_j) - T_j \geq f(q_i, \theta_j) - T_i.$$

Adding yields

$$f(q_j, \theta_j) - f(q_i, \theta_j) \geq f(q_j, \theta_i) - f(q_i, \theta_i).$$

Strict increasing differences rules out $q_j < q_i$. Hence $q_j \geq q_i$, and transfers are ordered by IC when quantities are strictly ordered.

Theorem 7.1 (SSCP and increasing differences for quasilinear contracts). *For $u(q, T, \theta) = f(q, \theta) - T$, strict single crossing in $((q, T), \theta)$ is equivalent to strictly increasing differences of f in (q, θ) .*

Proof. If f has strictly increasing differences and $q' > q$, then

$$f(q', \theta') - f(q, \theta') > f(q', \theta) - f(q, \theta) \quad (\theta' > \theta).$$

Rearrange to get the strict single-crossing implication for the utility difference between (q', T') and (q, T) . Conversely, apply strict single crossing to pairs with transfers chosen so that the lower type is exactly indifferent; the higher type must then strictly prefer the higher- q contract, which is exactly strict increasing differences. $\square$

7.2 Rothschild–Stiglitz insurance screening

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw accident-state wealth on the horizontal axis and no-accident wealth on the vertical axis; zero-profit lines fan out by risk type.*

Object. Consumers have initial wealth w and face loss ℓ . Type $\theta \in \{\theta_L, \theta_H\}$ is accident probability, with $0 < \theta_L < \theta_H < 1$. A contract (x, t) gives coverage $x \in [0, \ell]$ and premium t . Wealth is

$$w_N = w - t, \quad w_A = w - \ell + x - t.$$

Expected utility is

$$U_\theta(x, t) = (1 - \theta)u(w - t) + \theta u(w - \ell + x - t),$$

with $u' > 0, u'' < 0$.

Feasibility. Firms are risk neutral. Expected profit from type θ is

$$\pi_\theta(x, t) = t - \theta x.$$

A separating competitive candidate has zero profit by type:

$$t_L = \theta_L x_L, \quad t_H = \theta_H x_H.$$

Feasibility also includes type IC and participation relative to no insurance.

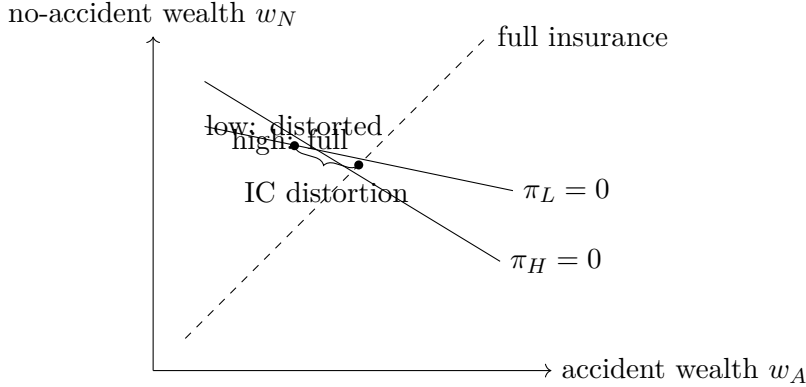
Optimality/Equilibrium. Under full information, each type buys full insurance $x = \ell$ at actuarially fair premium $t = \theta\ell$. Under asymmetric information, the high-risk type must not want the low-risk contract. The separating candidate is usually

$$(x_H, t_H) = (\ell, \theta_H \ell),$$

$$t_L = \theta_L x_L, \quad U_{\theta_H}(x_L, \theta_L x_L) = U_{\theta_H}(\ell, \theta_H \ell),$$

with $x_L < \ell$.

Verification. Verify in order: high type gets full insurance; low contract breaks even; high type's IC binds; low type prefers the low contract by single crossing; both participation constraints hold; and no entrant can offer a contract earning positive profit and attracting a profitable subset of types. The last step is why a Rothschild–Stiglitz equilibrium can fail to exist.



7.3 Credit-market screening

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw collateral on the horizontal axis and interest on the vertical axis with borrower indifference curves.*

Object. A loan contract is (r, x) , where r is interest and x is collateral. A borrower has insolvency probability $q \in \{q_\ell, q_h\}$, $q_\ell < q_h$. Borrower payoff is

$$B(q; r, x) = 1 - r(1 - q) - qx,$$

and lender payoff is

$$L(q; r, x) = r(1 - q) - qbx, \quad 0 < b < 1.$$

Feasibility. Zero-profit contracts satisfy

$$r = \frac{qb}{1 - q}x.$$

Type IC constraints require each borrower type to choose its intended contract. Because collateral is inefficient for the lender but differentially costly to types, it can screen.

Optimality/Equilibrium. With observable type, zero collateral is efficient and the break-even interest rate satisfies

$$r_i = \frac{q_i}{1 - q_i},$$

if the loan principal is normalized appropriately. With private type, a separating candidate typically gives the high-risk type its full-information contract and distorts the low-risk contract using collateral/interest so that the high-risk type does not imitate.

Verification. Check borrower IC, lender zero profit by contract, participation, and whether a deviating lender can cream-skim low-risk borrowers with a profitable contract. Draw the candidate in (x, r) space: zero-profit lines and indifference curves often prove uniqueness/nonexistence faster than algebra.

8 Verifiable disclosure and hard information

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a type ladder where each type can either reveal its rung or step into an ambiguous pooled box, but cannot claim another rung.*

8.1 Canonical verifiable-disclosure game

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw messages as subsets of the type space; a message is feasible only if it contains the true type.*

Object. Nature draws sender type $\theta \in \Theta$, observed by the sender. The sender chooses a verifiable message $m \in M(\theta)$. A common formalization is that m is an event/subset $m \subseteq \Theta$ with $\theta \in m$: the sender may disclose truthfully or partially, but cannot lie. The receiver observes m , forms posterior $\mu(\cdot | m)$ supported on m , and chooses action $a \in A$.

Feasibility. A sender strategy $s : \Theta \rightarrow M$ must satisfy $s(\theta) \in M(\theta)$. A receiver strategy $r : M \rightarrow A$ must be optimal given beliefs. Beliefs must assign zero probability to types inconsistent with the message, and on-path beliefs must satisfy Bayes' rule.

Optimality/Equilibrium. Receiver action is

$$r(m) \in \arg \max_{a \in A} \sum_{\theta \in m} \mu(\theta | m) u_R(a, \theta).$$

Sender optimality requires

$$u_S(r(s(\theta)), \theta) \geq u_S(r(m'), \theta) \quad \forall m' \in M(\theta).$$

When sender utility is increasing in receiver action and receiver action is increasing in posterior expected type, the highest type tends to reveal, then the next highest reveals, and so on: this is the unraveling logic.

Verification. To verify disclosure equilibrium, check message feasibility first. Then compute the receiver's posterior after every on-path message. Finally, ask whether a type inside an ambiguous pool can separate by sending a more precise feasible message that improves the receiver's action. If yes, the pooling disclosure cannot be a wPBE.

Theorem 8.1 (Simple unraveling theorem). *Let $\Theta = \{\theta_1 < \dots < \theta_n\}$. Suppose each type can send any message m containing its true type, including the singleton $\{\theta_i\}$. Suppose the receiver's optimal action is increasing in $\mathbb{E}[\theta | m]$ and every sender type weakly prefers higher receiver actions, with strict preference when actions differ. Then every equilibrium outcome is fully revealing.*

Proof. If an equilibrium has a non-singleton on-path pool m , let θ_k be the highest type in the pool. By sending the singleton $\{\theta_k\}$, type θ_k induces posterior mean θ_k , which is strictly higher than the posterior mean of the nondegenerate pool. The receiver's action is strictly higher, so type θ_k strictly gains. Hence no non-singleton pool can be on path. Iterating leaves only singleton disclosure. $\square$

8.2 Salesperson quality disclosure

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw the buyer's demand curve shifting right as posterior expected quality rises.*

Object. Quality is $\theta \in \{\theta_L, \theta_H\}$, $0 < \theta_L < \theta_H$ and $\theta_L > p$. The salesperson observes θ ; the buyer does not. Type t can send m_t (“quality is t ”) or m_0 (“quality is either L or H ”). The buyer chooses quantity $q \geq 0$ at price p with expected utility

$$\mathbb{E}[\theta \mid m]q - \frac{1}{2}q^2 - pq.$$

The salesperson receives $R(q)$, with R strictly increasing.

Feasibility. Message feasibility is

$$M_L = \{m_L, m_0\}, \quad M_H = \{m_H, m_0\}.$$

If the buyer’s posterior probability of low quality is π , then

$$\mathbb{E}[\theta \mid \pi] = \pi\theta_L + (1 - \pi)\theta_H.$$

Because $\theta_L > p$, the buyer’s demand is interior:

$$q(\pi) = \mathbb{E}[\theta \mid \pi] - p.$$

Optimality/Equilibrium. The high type can always send m_H , which induces $q_H = \theta_H - p$. Since no other belief can induce a higher expected quality, the high type must send m_H in every wPBE, except for irrelevant indifference cases that would make the low type want to mimic. The low type cannot send m_H . It can send either m_L or m_0 . If high sends m_H , then either low message reveals low quality: m_L directly, or m_0 by Bayes’ rule because only the low type uses it.

Verification. The wPBE outcomes are fully revealing. One equilibrium has $L \mapsto m_L$, $H \mapsto m_H$, with off-path belief after m_0 assigning probability one to L to deter low deviation. Another has $L \mapsto m_0$, $H \mapsto m_H$; then Bayes’ rule after m_0 assigns probability one to L . There is no pooling wPBE on m_0 because high would deviate to m_H and induce $q_H > q(1/2)$.

Welfare comparison with no communication. With no communication, the buyer chooses

$$q_0 = \frac{\theta_L + \theta_H}{2} - p = \frac{q_L + q_H}{2}.$$

The buyer’s ex ante gross surplus from an optimal quantity under information I is $\frac{1}{2}(\mathbb{E}[\theta \mid I] - p)^2$. Hence the buyer strictly prefers disclosure:

$$\frac{1}{4}(q_L^2 + q_H^2) > \frac{1}{2}\left(\frac{q_L + q_H}{2}\right)^2$$

when $q_L \neq q_H$. The salesperson compares

$$\frac{1}{2}R(q_L) + \frac{1}{2}R(q_H) \quad \text{with} \quad R\left(\frac{q_L + q_H}{2}\right).$$

Thus no communication is better for the salesperson if R is sufficiently concave, worse if R is sufficiently convex, and equal if R is affine.

8.3 Disclosure in Bayesian Cournot

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw two output best-reply diagrams: pooled-cost belief and disclosed low-cost belief.*

Object. Firm 2 has private unit cost $k + \varepsilon$, where $\varepsilon \in \{-e, e\}$ with equal probability and mean zero. Firm 1's unit cost is k . Demand is $P(Q) = a - Q$. Before choosing outputs, firm 2 can disclose its cost verifiably or remain silent. If it discloses, firm 1 learns ε ; if it is silent, firm 1 updates from equilibrium behavior.

Feasibility. A disclosure strategy for firm 2 maps each type ε to D or C . Firm 1's output strategy maps observed disclosure/silence to an output, possibly conditional on beliefs after silence. After disclosure, beliefs are degenerate; after silence, Bayes' rule applies if silence is on path.

Optimality/Equilibrium. In the pooling-on-silence candidate, firm 1's belief after silence is the prior. The Bayesian Cournot equilibrium is

$$q_1^P = \frac{a - k}{3}, \quad q_2^P(\varepsilon) = \frac{a - k}{3} - \frac{\varepsilon}{2}.$$

In the complete-information subgame after disclosure of ε ,

$$q_1^D(\varepsilon) = \frac{a - k + \varepsilon}{3}, \quad q_2^D(\varepsilon) = \frac{a - k - 2\varepsilon}{3}.$$

Because Cournot profit equals own equilibrium output squared in the interior linear-demand case, type ε 's profit under silence is $[q_2^P(\varepsilon)]^2$ and under disclosure is $[q_2^D(\varepsilon)]^2$.

Verification. The low-cost type has $\varepsilon = -e$. It compares

$$q_2^P(-e) = \frac{a - k}{3} + \frac{e}{2} \quad \text{with} \quad q_2^D(-e) = \frac{a - k}{3} + \frac{2e}{3}.$$

Disclosure gives strictly higher output and profit. Hence there is no wPBE in which both types remain silent. Moreover, any pure-strategy equilibrium in which firm 1 fails to learn cost would require both types to send the same nonrevealing message; the low-cost type would deviate to disclosure. Therefore firm 1 learns the cost in every pure-strategy wPBE.

9 Graphical tools

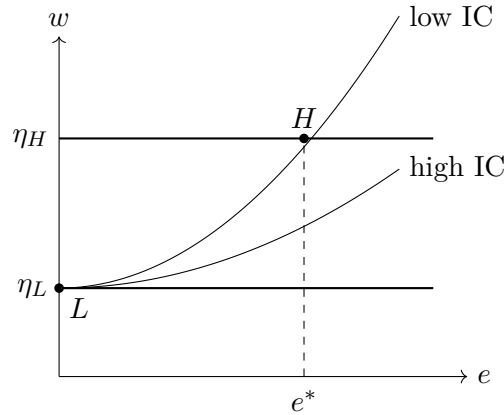
Visual intuition hook. *Pair with Visual Intuition Curriculum: draw each graph on one index card: signaling, insurance, disclosure, and best replies.*

9.1 Wage-education diagrams

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw high-type and low-type indifference curves through the same wage-education point and label the flatter high-type curve. In Spence diagrams, the vertical axis is wage and the horizontal axis is education. Indifference curves satisfy*

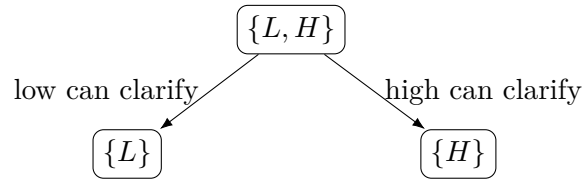
$$w - c(e, \eta) = \bar{U}.$$

Their slope is $c_e(e, \eta)$. Single crossing means the high-productivity type has lower marginal cost, so its indifference curve is flatter. A separating equilibrium puts the low type at $(0, \eta_L)$ and the high type at (e^*, η_H) with e^* in the IC interval.



9.2 Disclosure lattice

Visual intuition hook. *Pair with Visual Intuition Curriculum:* draw a set-inclusion lattice of messages: singleton truths at the bottom, pooled ambiguous messages above. For verifiable disclosure, draw feasible messages as sets of possible types. A type can move only to messages containing its true type. The unraveling proof starts at a non-singleton message and asks whether the highest type in that set can move down the lattice to a singleton truth.



9.3 Insurance state-space diagrams

Visual intuition hook. *Pair with Visual Intuition Curriculum:* draw the 45-degree line as full insurance and zero-profit lines as budget lines through endowment. The accident-state diagram is often faster than algebra. Full insurance lies on $w_A = w_N$. A type- θ zero-profit line has slope $-(1 - \theta)/\theta$ in (w_A, w_N) space. High-risk zero-profit lines are flatter/steeper depending on axis convention; label them carefully. The low-risk contract is distorted away from full insurance to relax the high-risk IC constraint.

9.4 Bayesian best-reply diagrams

Visual intuition hook. *Pair with Visual Intuition Curriculum:* draw one reaction curve for the uninformed player and one reaction curve for each informed type. In Bayesian Cournot, draw firm 1's type-contingent best replies and firm 2's expected best reply. The uninformed player's best reply depends on expected output, while each informed type's best reply depends on its own cost. Disclosure shifts the opponent's belief and therefore shifts the opponent's best reply.

10 Proof-writing templates

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a checklist next to each theorem: assumptions, claim, contradiction or inequalities, conclusion.*

10.1 Template: prove a candidate is BNE

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a table with one row per type and one column per deviation.*

Proof-writing template. Object. “A pure strategy for player i is a map $s_i : T_i \rightarrow A_i$. Consider candidate s^* .”

Feasibility. “The candidate is feasible because $s_i^*(t_i) \in A_i$ for every type. Conditional beliefs are induced by the common prior.”

Optimality/Equilibrium. “For type t_i , expected payoff from the prescribed action is $\dots$; from deviation a_i is $\dots$. The prescribed action is optimal iff inequality $\dots$ holds.”

Verification. “The inequalities hold for every type of every player. Therefore s^* is a BNE. To show all BNE, the best-response table exhausts all pure strategies.”

10.2 Template: prove a candidate is WPBE

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a receiver belief at every message and a deviation arrow for every sender type.*

Proof-writing template. Object. “An assessment is (s_1, s_2, μ) .”

Feasibility. “The sender messages are type-feasible. Beliefs are distributions. At on-path messages, Bayes’ rule gives $\mu(\cdot \mid m) = \dots$.”

Optimality/Equilibrium. “Given μ , the receiver’s action $s_2(m)$ maximizes posterior expected payoff. Given s_2 , each sender type weakly prefers its prescribed message to each feasible deviation.”

Verification. “All information sets and all deviations have been checked; therefore the assessment is a WPBE. If an off-path belief is used, state exactly where it is used.”

10.3 Template: prove monotonicity from single crossing

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw two contracts and two indifference curves, then mark the crossing contradiction.*

Proof-writing template. Let $\theta_H > \theta_L$. Suppose type H chooses x_H and type L chooses x_L . IC gives

$$u(x_H, \theta_H) \geq u(x_L, \theta_H), \quad u(x_L, \theta_L) \geq u(x_H, \theta_L).$$

If $x_H < x_L$, strict single crossing says the lower type’s weak preference for x_L over x_H implies the higher type’s strict preference for x_L over x_H , contradicting high-type IC. Therefore $x_H \geq x_L$.

10.4 Template: prove no pooling in disclosure

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a pooled message and an arrow from the best type to its singleton message.*

Proof-writing template. Assume a nonrevealing message m is used by at least two types. Let θ^* be the type in the pool that most benefits from receiver optimism, usually the highest quality or lowest cost. Because disclosure is verifiable, θ^* can send a more precise feasible message m^* that excludes worse types. Bayes' rule then makes the receiver more favorable to θ^* . Since θ^* strictly prefers the induced action, it profitably deviates. Contradiction.

11 Qual-style drills with solution skeletons

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw each drill as a mini-tree or mini-payoff table before writing equations.*

11.1 Drill 1: enumerate pure BNE in a two-type game

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw four pure strategies as corners of a square: YY, YN, NY, NN . Use the R&D race calculation in Section 4.3. The answer must contain: (i) pure strategy set; (ii) expected payoff to each type from entry against each opponent strategy; (iii) best-response correspondence; (iv) fixed points; (v) robustness comment. The pure BNE are (YY, YY) , (YN, YN) , and (NN, NN) .*

11.2 Drill 2: find a separating Spence equilibrium

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw the low type at zero education and the high type at the lowest education that deters the low type. Given $c_L(e) > c_H(e)$ and productivities $\eta_H > \eta_L$, write the separating IC interval:*

$$c_L(e) \geq \eta_H - \eta_L \geq c_H(e).$$

The least-cost separating signal binds the low type's constraint. Verify high type's constraint, receiver zero profit, Bayes' rule, and off-path beliefs.

11.3 Drill 3: determine whether pooling on an ambiguous disclosure is wPBE

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw the ambiguous message as a pool and a singleton truthful message outside the pool. Ask: can one type send a truthful singleton that improves the receiver action? In quality-sales problems, high quality can reveal H , so pooling on m_0 fails. In low-cost Cournot disclosure, low cost can disclose, inducing a more favorable complete-information rival output, so pooling on silence fails.*

11.4 Drill 4: write a screening menu problem

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a menu with two dots and label IC arrows in both directions. For two types, write*

$$\max_{(q_L, T_L), (q_H, T_H)} p_L T_L + p_H T_H$$

subject to

$$\begin{aligned} f(q_L, \theta_L) - T_L &\geq 0, & f(q_H, \theta_H) - T_H &\geq 0, \\ f(q_L, \theta_L) - T_L &\geq f(q_H, \theta_L) - T_H, \\ f(q_H, \theta_H) - T_H &\geq f(q_L, \theta_H) - T_L. \end{aligned}$$

Then use single crossing to identify monotonicity and likely binding constraints.

12 Common pitfalls and fixes

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a red X over algebra-only answers and a green check over strategy/belief/IC tables.*

12.1 Pitfall: forgetting receiver strategies off path

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw an unreached message with a receiver action still written below it.* A WPBE strategy must specify what the receiver does after every message, not only after messages observed in equilibrium. Off-path receiver actions require beliefs and sequential rationality.

12.2 Pitfall: treating BNE as state-by-state Nash equilibrium

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw two complete-information games and cross out the mistaken idea that the Bayesian game is just their average.* A Bayesian game is not a list of complete-information games. A player who does not observe the state cannot condition on it. Strategies condition only on own information.

12.3 Pitfall: using FOCs before proving concavity or monotonicity

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a nonconcave objective with multiple local peaks and mark why an FOC can mislead.* When action sets are continua, state why FOCs are sufficient: concavity, single crossing plus monotone methods, or direct global comparison. Otherwise use inequalities.

12.4 Pitfall: saying “pooling” when the receiver still learns

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a message m_0 used only by low type and label it as revealing by Bayes’ rule.* A message may look vague but be fully revealing in equilibrium. If only one type uses the vague message, Bayes’ rule identifies that type.

13 One-page formula sheet

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a single page divided into BNE, WPBE, Spence, screening, and disclosure boxes.*

13.1 Bayesian games

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a type-contingent strategy as a function arrow $T_i \rightarrow A_i$.*

$$\text{BNE: } s_i^*(t_i) \in \arg \max_{a_i \in A_i} \sum_{t_{-i}} u_i(t_i, t_{-i}, a_i, s_{-i}^*(t_{-i})) \beta_i(t_{-i} \mid t_i).$$

13.2 WPBE

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw Bayes' numerator over denominator at a receiver information set.*

$$\mu(x \mid h) = \frac{P_\sigma(x)}{P_\sigma(h)} \quad \text{if } P_\sigma(h) > 0.$$

$$\sigma_i \in \arg \max_{\sigma'_i} E[u_i \mid h, \mu_i, \sigma'_i, \sigma_{-i}] \quad \forall h \in \mathcal{H}_i.$$

13.3 Spence two-type separation

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw the IC interval as a segment between $c_L^{-1}(\Delta\eta)$ and $c_H^{-1}(\Delta\eta)$.*

$$\Delta\eta = \eta_H - \eta_L, \quad c_L(e^*) \geq \Delta\eta \geq c_H(e^*).$$

Least-cost separating: $c_L(e^*) = \Delta\eta$.

13.4 Screening monotonicity

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw higher type choosing weakly higher quantity. Increasing differences $\Rightarrow$ IC allocation monotone:*

$$\theta_H > \theta_L \quad \Rightarrow \quad q_H \geq q_L.$$

13.5 Disclosure unraveling

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw the top type leaving a pool by truthful singleton disclosure. Pooling fails if a type can send a feasible, more precise truth that induces a strictly better receiver action.*

14 References for further study

Visual intuition hook. *Pair with Visual Intuition Curriculum: draw a reading stack: notes first, Maschler–Solan–Zamir second, classic papers third, past quals last.*

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